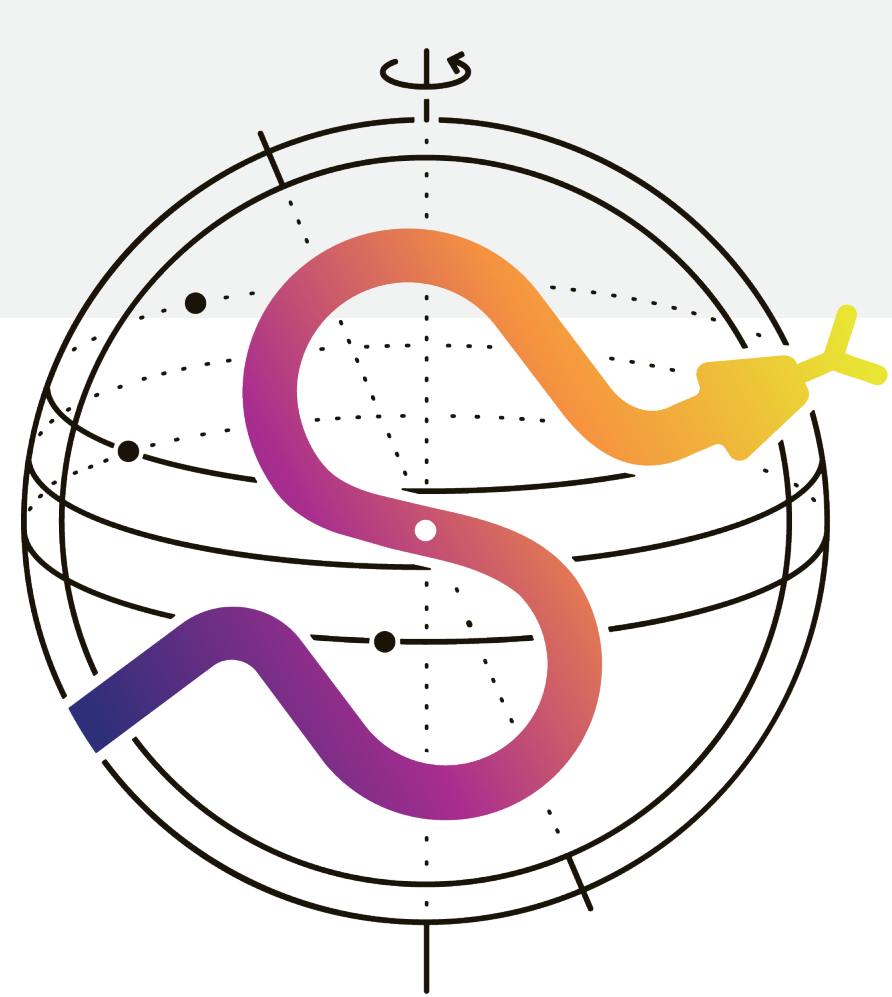




Redist: a python tool for model-agnostics binned-likelihood fits in High Energy Physics

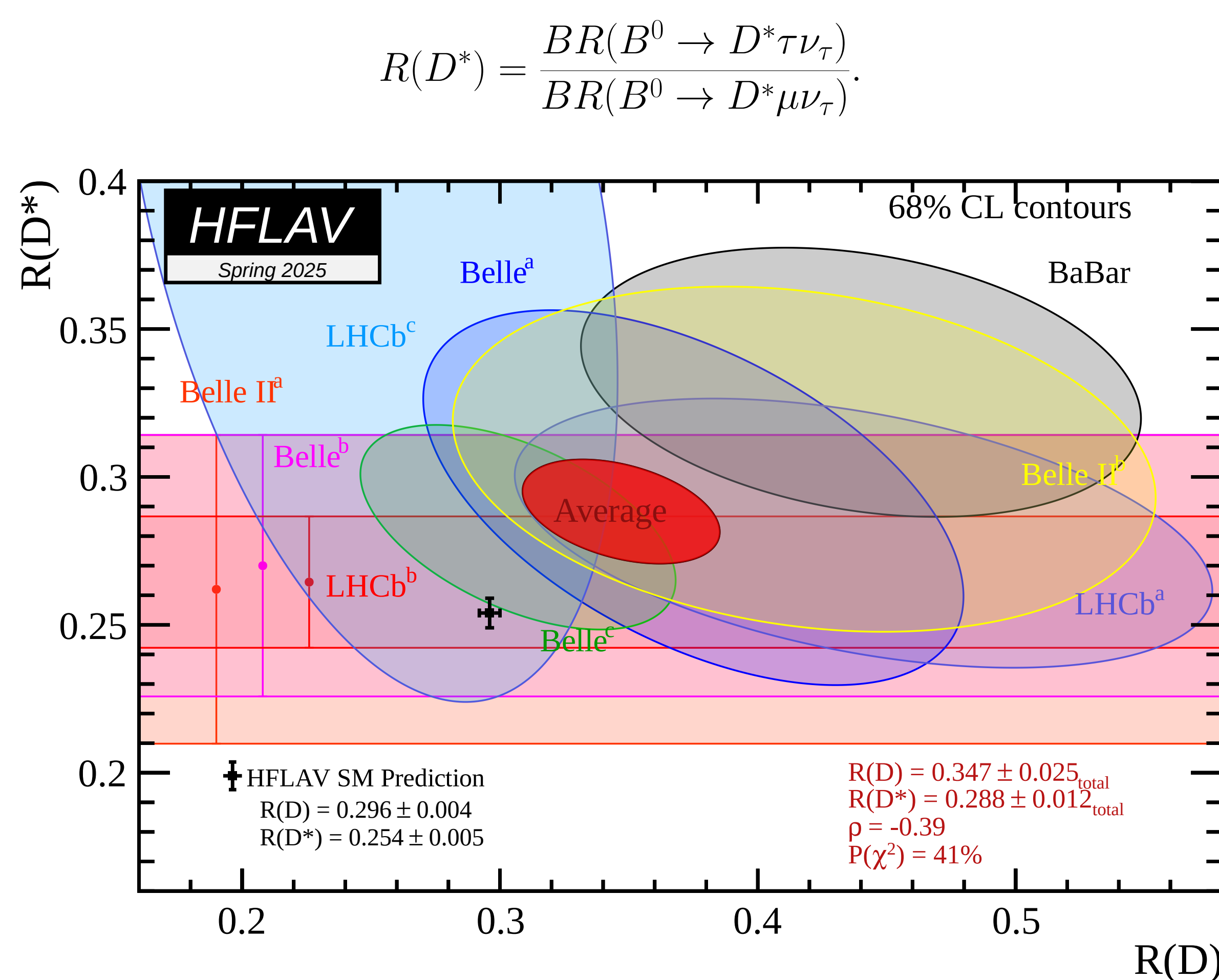
The problem of reinterpretation of data in HEP

HEP analyses rely on **theoretical assumptions**:

- **Standard Model** predictions (to test the Standard Model);
- **underlying kinematic** assumptions in the **simulations** of the processes.

Is it possible to build model-agnostic analyses?

The current $R(D)$ and $R(D^*)$ measurements report a **discrepancy with the SM** (fig):



We observe **hints of Beyond the Standard Model (BSM) effects**:

- example: a **new particle** not included in the SM.

We can measure BSM contributions directly.

How can we reinterpret the data under different assumptions?

Can we combine datasets? Do we gain sensitivity?

J.Albrecht¹, F.Bernlochner², **Marco Colonna**¹, L.Gärtner³, A.Mathad⁴, and B.Mitreska¹

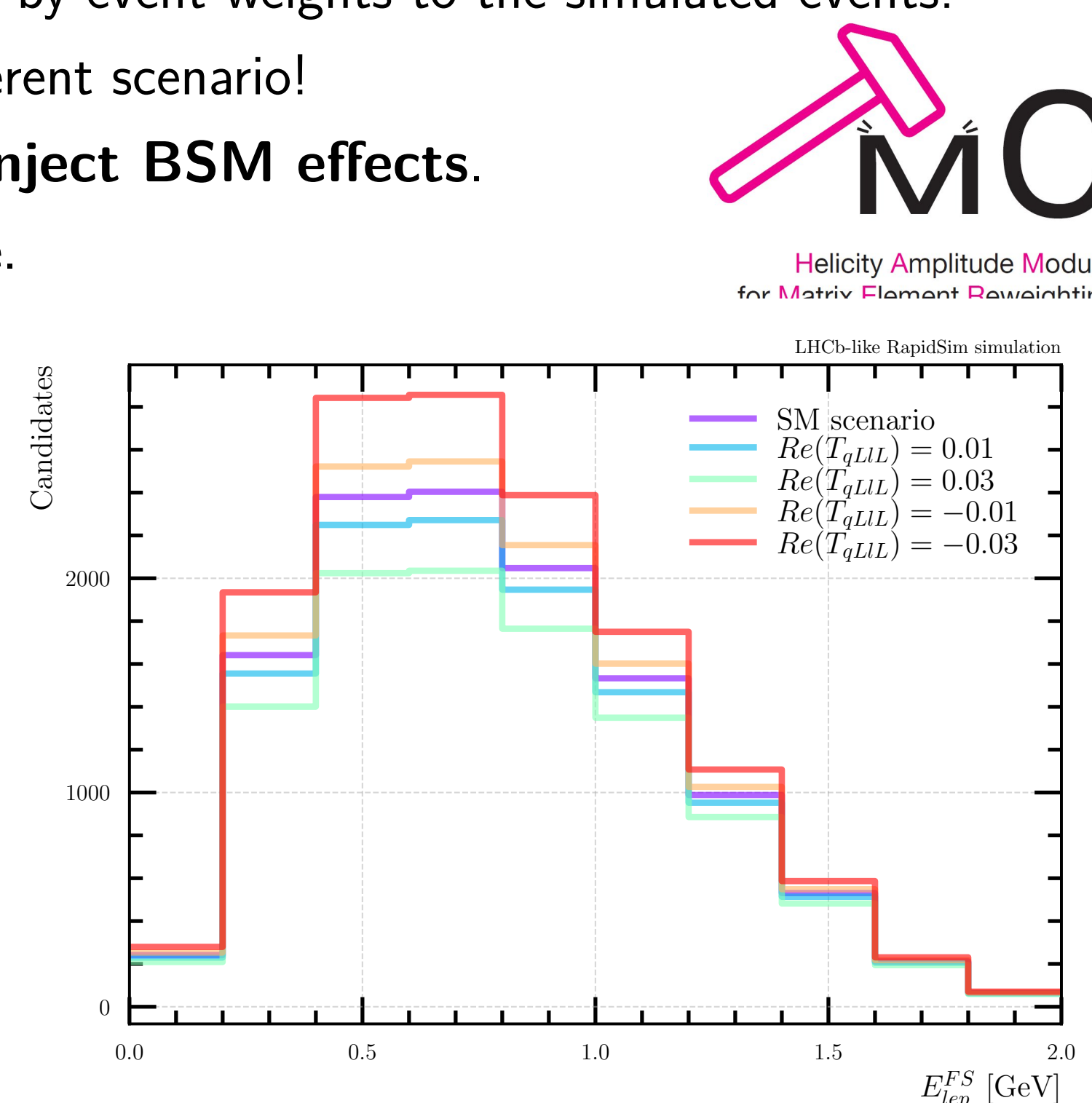
¹TU Dortmund University, ²University of Bonn, ³LMU Munich University, ⁴CERN

HAMMER as theoretical backend

The **HAMMER** package applies event-by-event weights to the simulated events:

- **No need to oversimulate** each different scenario!
- **Change Form Factors** (QCD) and **inject BSM effects**.
- **Model-agnostic** analyses are possible.

Current	WC Name	4-Fermi/(i2√2V _{cb} G _F)
Standard Model	SM	$[c\gamma^\mu P_L b][\ell\gamma^\mu P_L \nu_\tau]$
Vector	$V_{-q}LIL$	$[e\chi_L^V \gamma^\mu P_L b][\ell\lambda_L^V \gamma^\mu P_L \nu_\tau]$
	$V_{-q}RIL$	$[e\chi_R^V \gamma^\mu P_R b][\ell\lambda_L^V \gamma^\mu P_L \nu_\tau]$
Scalar	$S_{-q}LIL$	$[e\chi_L^S P_L b][\ell\lambda_L^S P_L \nu_\tau]$
	$S_{-q}RIL$	$[e\chi_R^S P_R b][\ell\lambda_L^S P_L \nu_\tau]$
Tensor	$T_{-q}LIL$	$[e\chi_L^T \sigma^{\mu\nu} P_L b][\ell\lambda_L^T \sigma_{\mu\nu} P_L \nu_\tau]$
	$T_{-q}RIL$	$[e\chi_R^T \sigma^{\mu\nu} P_R b][\ell\lambda_L^T \sigma_{\mu\nu} P_L \nu_\tau]$



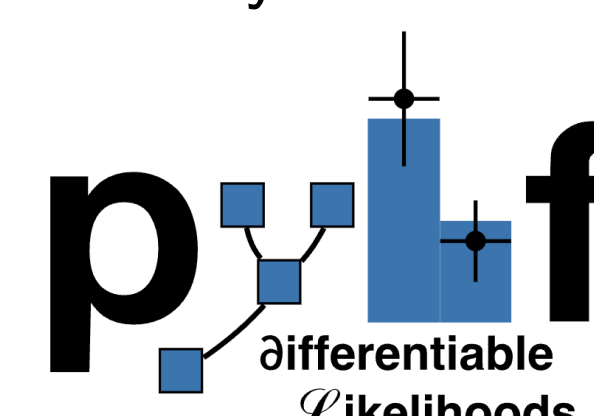
Pyhf as fitting interface

The **pyhf** implements **HistFactory** in **Python**. The **likelihood** is made by:

$$L_{data}(n; \eta, \chi) = \prod_{c \in \text{channel}} \prod_{b \in \text{bins}} \text{Pois}(n_{cb} | \nu_{cb}(\eta, \chi)) \quad (\text{Data})$$

$$C_{exp} = \prod_{\chi \in \chi_{exp}} c_\chi(a_\chi | \chi) \quad (\text{Experimental constraints})$$

$$C_{th} = \prod_{\chi \in \chi_{th}} c_\chi(a_\chi | \chi) \quad (\text{Theoretical constraints})$$



The **rates** are varied using a set of **additive** (Δ) and **multiplicative** (k) **modifiers**:

$$\nu_{cb} = \sum_{s \in \text{samples}} \prod_{k \in k} k_{scb}(\eta, \chi) \times \left(\nu_{scb}^0(\eta, \chi) + \sum_{\Delta \in \Delta} \Delta_{scb}(\eta, \chi) \right)$$

How can we include the HAMMER parameters in pyhf?

Can we extract their values from a fit?

The Redist custom modifier for pyhf

The **Redist** package introduces a **pyhf** **custom modifier**: it updates the model based on new theoretical assumptions.

To define a custom modifier, we need:

- a set of parameters (BSM and Form Factors);
- a baseline hypothesis function (SM HAMMER histograms);
- an alternative hypothesis function (BSM HAMMER histograms).



How the custom modifier works

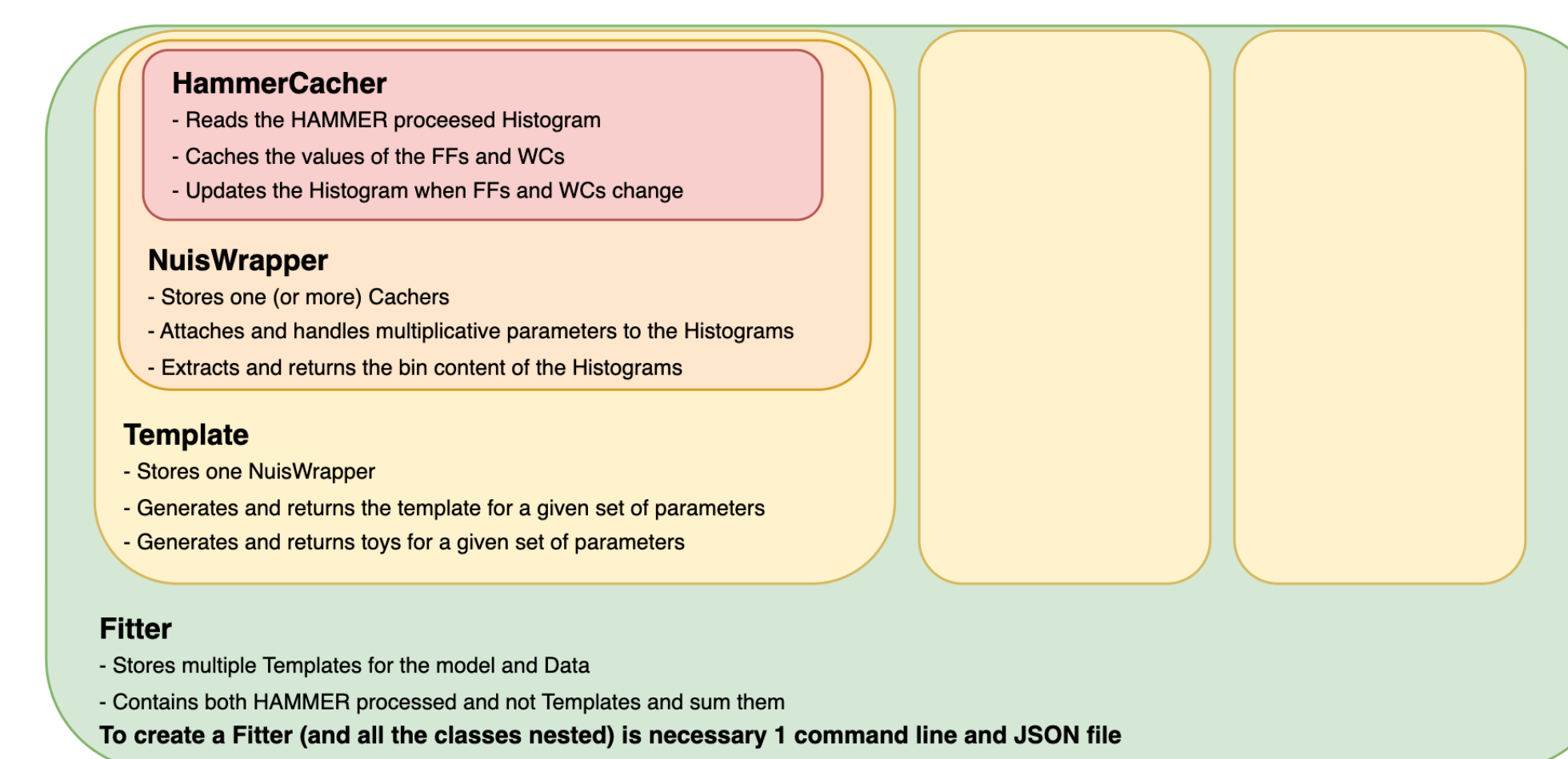
- A baseline distribution ($M(\vec{p}_0)$) is stored and updated with a function of the new parameters ($f(\vec{p})$):

$$M(\vec{p}) = M(\vec{p}_0) \times \frac{f(\vec{p})}{f_0(\vec{p}_0)}$$

- if $M(\vec{p}_0) = f_0(\vec{p}_0)$ than $M(\vec{p}) = f(\vec{p})$ with $f(\vec{p})$ being arbitrarily complex:
 - the parameters in f will be **handled by the theoretical backend**;
 - the **method is general** and fits any backend (ex: EOS, and **now HAMMER!**).

The **Redist-HAMMER** interface has been built to use HAMMER within pyhf:

- **nested class structure** and the principle of **single responsibility**.



Development of Data Reinterpretation in python

As data reinterpretation is becoming a **discussed topic** in HEP:

- Redist offers a setup working with **different theoretical backends**.

Fields outside HEP benefits of the Redist for any binned-log-likelihood fit:

- **popular and flexible pyhf** model building;
- handling of **complex degrees of freedom** in the fit.



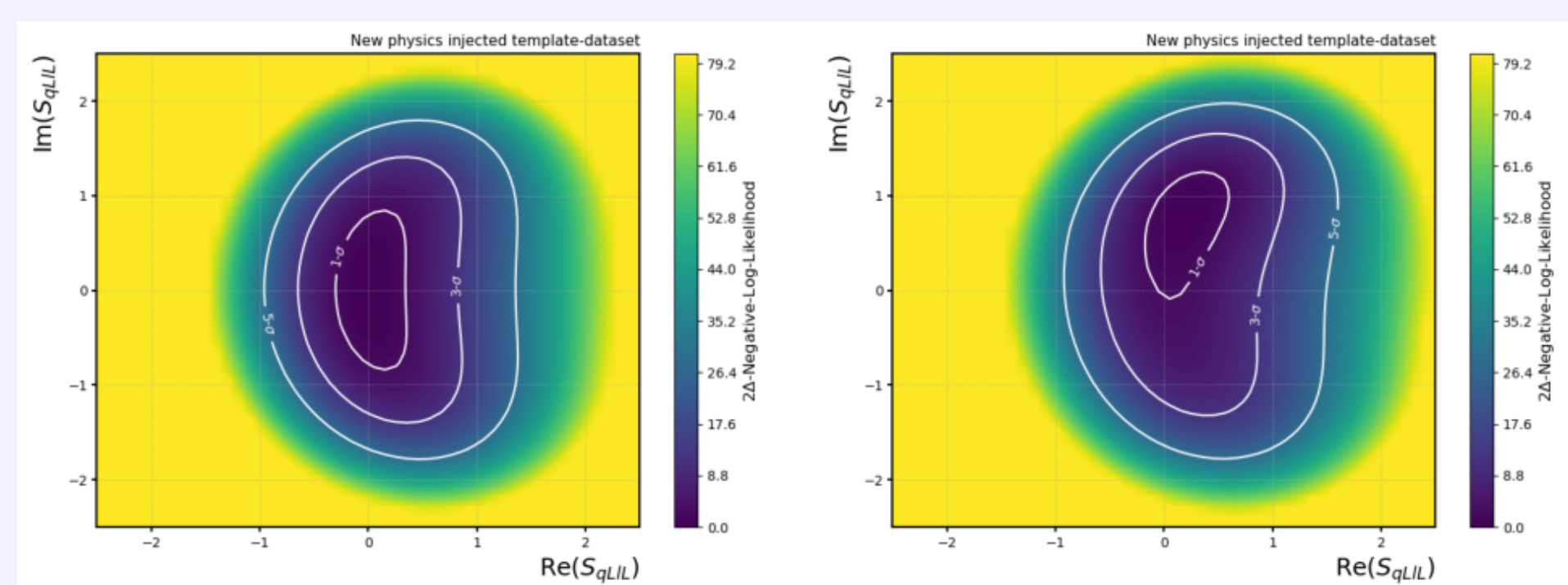
Our GitHub

Achievements, ongoing progress and future prospects on HEP data reinterpretation in Python

Fit an NP parameter and draw the confidence intervals in phase space

The $B^0 \rightarrow D^* \tau \nu_\tau$ sample has been **HAMMER-processed**:

- A **sensitivity study** on the Wilson coefficients has been performed with a **template phase-space scan**;
- The Δ -NLL for each phase-space point provides the **confidence intervals, accounting for parameter correlations**.

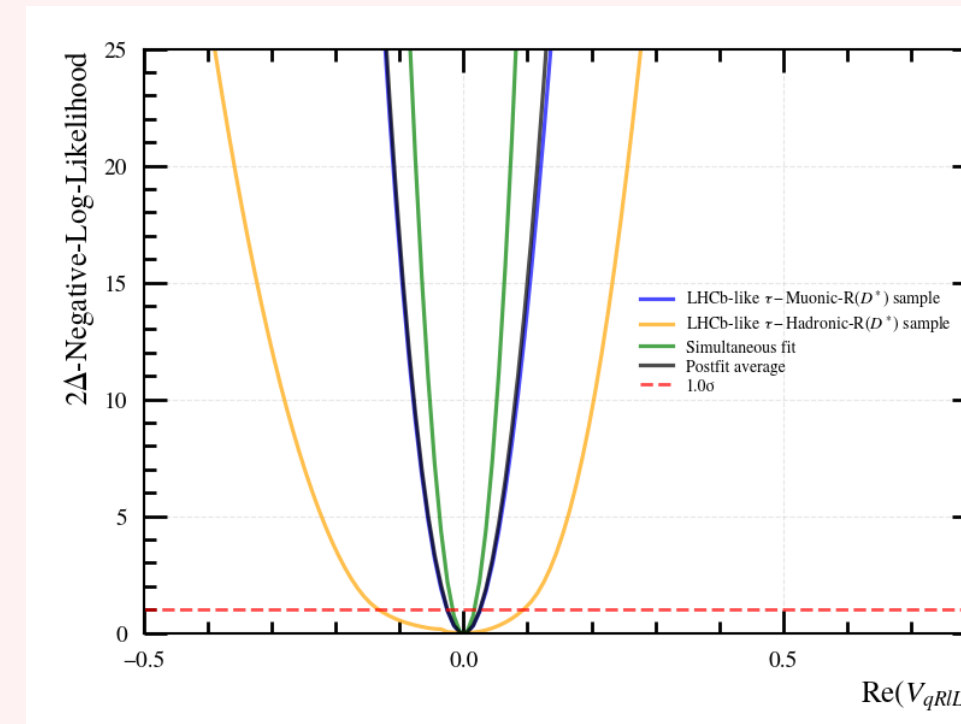


- **Different modes** exhibit **different confidence level shapes**.
- Studies with **NP-injected samples** are now possible at the **major HEP experiments**: LHCb, Belle II, CMS and Atlas!

Combine different datasets in a simultaneous NP fit

The combination of two $R(D^*)$ LHCb-like measurements:

- Two models have been built to simultaneously fit the pseudo-data;
- The **advantages of the simultaneous fit** over a post-fit average have been established:
 - **bias correction**;
 - **consistency** of the shared parameters;
 - **increased sensitivity**.
- **Different modes** show **different sensitivities to be combined**.



Redist-HAMMER provides a **coherent framework** to combine measurements from **different samples or experiments**.

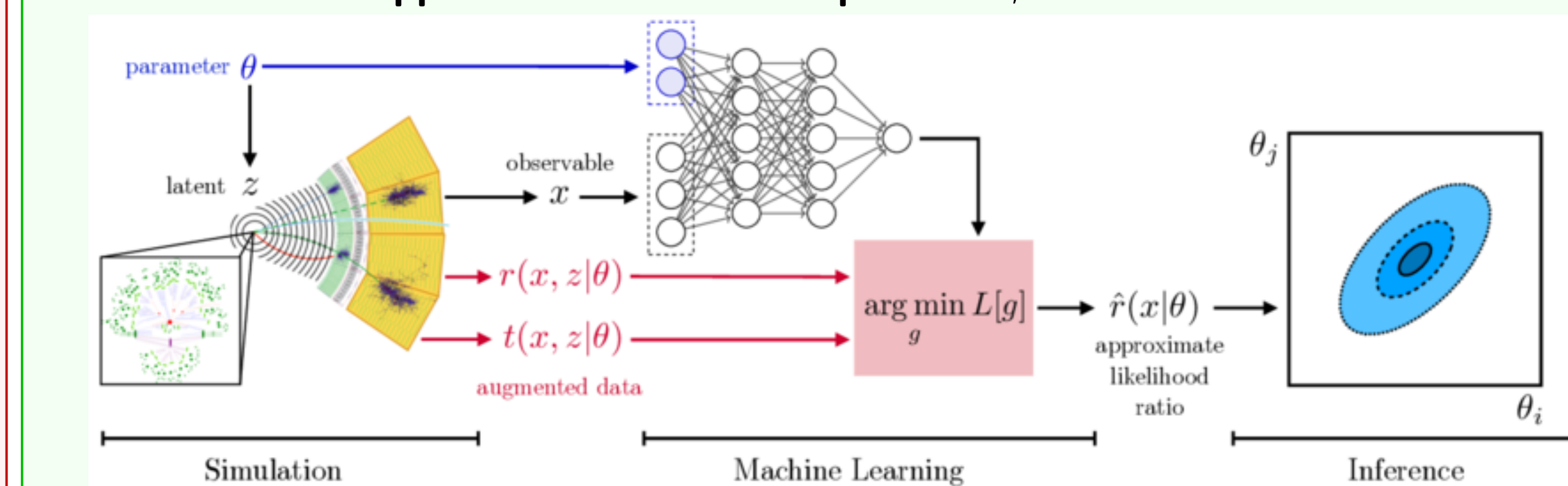
Current efforts on the **combination of $R(D^*)$ measurements** are based on **Redist-HAMMER**.

Other analyses are using **Redist-HAMMER** to fit NP parameters in Python.

Build baseline fits to compare with other NP techniques and develop them

A technique of **simulation-based inference (SBI)** is under development at ROOT:

- It is a **different approach** to data reinterpretation;



- **Redist-HAMMER** is used as a **baseline fit** to compare and quality-check the SBI.